



How to fix all, or at least a few, gaps and how to stand out with good proof and accuracy. Frame the system architecture like that

Now I have everything needed for a fully grounded, literature-backed architecture. Here is the complete system design.

EcoSynth — Differentiated Architecture

The core research position is this: **existing systems fix one gap in isolation**. AizynthFinder does multi-step planning but has no green scoring. RetroExplainer adds interpretability but only for single-step prediction. Current LLM-RAG drug discovery work grounds factual retrieval but doesn't use RAG as a chemistry constraint. **EcoSynth is the first system to close Gaps 2, 3, and 5 simultaneously in a single cohesive pipeline.** [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#)

The Five Modules (Each Closes a Specific Gap)

Module 1 — Constraint-Injected RAG Engine

Closes: Gap 3 (LLM hallucination)

This is architecturally the most critical innovation. Current RAG systems retrieve context and *suggest* it to the LLM. EcoSynth uses retrieval as **hard reaction constraints** — the LLM is only allowed to suggest disconnections that have structural precedent in the indexed reaction corpus. [\[4\]](#) [\[5\]](#)

How it works:

1. User inputs target SMILES
2. RDKit generates a 2048-bit Morgan fingerprint (radius=2)
3. MiniLM-L6-v2 encodes the textual query ("synthesise aspirin via esterification")
4. Both fingerprint + text embedding are concatenated into a **hybrid query vector** and sent to ChromaDB
5. Top-5 retrieved reactions are formatted as: "Allowed disconnections: [SMARTS patterns]" and injected into the Gemma 3 1B system prompt as *constraints*, not suggestions
6. The LLM generates routes only within that bounded chemical space

Why this stands out: The 2025 ACS JCIM study showed LLM hallucination produces 62.34 RMSE on molecular properties unconstrained, vs 11.76 with RAG grounding. EcoSynth's constraint injection goes further — it structurally *prevents* out-of-distribution SMILES from being generated rather than correcting them post-hoc.^[5]

Module 2 — SMILES Validity Firewall

Closes: Gap 3 (validation layer after generation)

A two-stage post-generation filter that no current open system has as an explicit architectural component:^[6] ^[5]

- **Stage 1 (Deterministic):** RDKit `Chem.MolFromSmiles()` — binary pass/fail in <1ms. Catches syntactically invalid SMILES
- **Stage 2 (Learned):** Fine-tuned ChemBERTa-77M binary classifier on a dataset of (valid SMILES, hallucinated SMILES) pairs. Catches *syntactically valid but chemically implausible* structures — e.g., pentavalent carbon, impossible stereocentres

Training the classifier (doable on Colab free tier):

- Positive class: 50K valid SMILES from PubChem
- Negative class: 50K corrupted SMILES generated by random SMARTS mutation + known LLM failure modes collected from the ACS JCIM hallucination study^[5]
- Fine-tune ChemBERTa-77M for binary classification (2 epochs, ~45 min on Colab T4)^[7] ^[8]
- Export weights → load in CPU inference mode in FastAPI (~180MB RAM)

If a generated route contains *any* intermediate that fails Stage 2, the system flags it, explains why (which structural feature caused rejection), and re-queries Gemma with a tightened constraint set. **This is the explainability loop.**

Module 3 — Unified Multi-Criteria Green Scorer

Closes: Gap 2 (no consensus green metric)

This directly responds to the 2025 ChemSusChem critique that "heterogeneous metrics abound with no consensus". EcoSynth computes five metrics and fuses them with a **Random Forest-learned weight vector**:^[9]

$$G_{score} = \sum_{i=1}^5 w_i \cdot \hat{m}_i$$

Metric	Symbol	Source	Computation
E-factor	$\hat{m}_1$	Sheldon 1992	(waste mass) / (product mass), computed from RDKit atom counts
Atom Economy	$\hat{m}_2$	Trost 1991	(MW product) / (sum MW all reactants) × 100

Metric	Symbol	Source	Computation
Process Mass Intensity	$\hat{m}_3$	ACS GCI PMI Calculator [10]	(total mass in) / (mass product)
CHEM21 Solvent Score	$\hat{m}_4$	CHEM21 Guide [11] [12]	Safety + Health + Environment scores (1–10 per category), looked up from local solvent table
Step Count Penalty	$\hat{m}_5$	Novel	Exponential decay: $e^{-0.2n}$ where n = number of steps

All five are normalised to before weighting. The w_i weights are **learned** by training a Random Forest regressor on the ACS GCI Pharmaceutical Roundtable's published benchmarked reaction dataset — mapping known reactions to expert-assigned greenness rankings. This produces an empirically grounded scoring model rather than a hand-tuned formula, which is the specific criticism levelled at all prior systems. [\[13\]](#) [\[14\]](#) [\[9\]](#)

The CHEM21 solvent score is particularly important — the guide unified four major pharma solvent guides (GSK, ACS GCI, Pfizer, Sanofi) into one ranking, making it the best available proxy for industrial greenness. No current AI retrosynthesis tool uses it computationally. [\[11\]](#)

Module 4 — AizynthFinder Integration Layer

Closes: Gap 1 (multi-step planning)

Rather than building a tree search from scratch, EcoSynth wraps AizynthFinder (AstraZeneca/MolecularAI, open source) as its multi-step planner. AizynthFinder uses Monte Carlo Tree Search (MCTS) guided by a neural network policy trained on USPTO templates. This is state-of-the-art for multi-step planning and already open-source. [\[2\]](#) [\[1\]](#)

EcoSynth's contribution is not the tree search itself — it is what sits on top of it:

- AizynthFinder generates candidate multi-step routes (as a JSON reaction tree)
- Each route is passed through the Validity Firewall (Module 2)
- Each surviving route is scored by the Green Scorer (Module 3)
- The LLM (Module 1) generates natural-language *explanations* for why each route was ranked the way it was, grounded in the retrieved reaction precedents

This is a **novel orchestration** of an existing planning engine with green scoring and LLM explanation — no current system does this. [\[15\]](#) [\[16\]](#)

Module 5 — Human-in-the-Loop Steering Interface

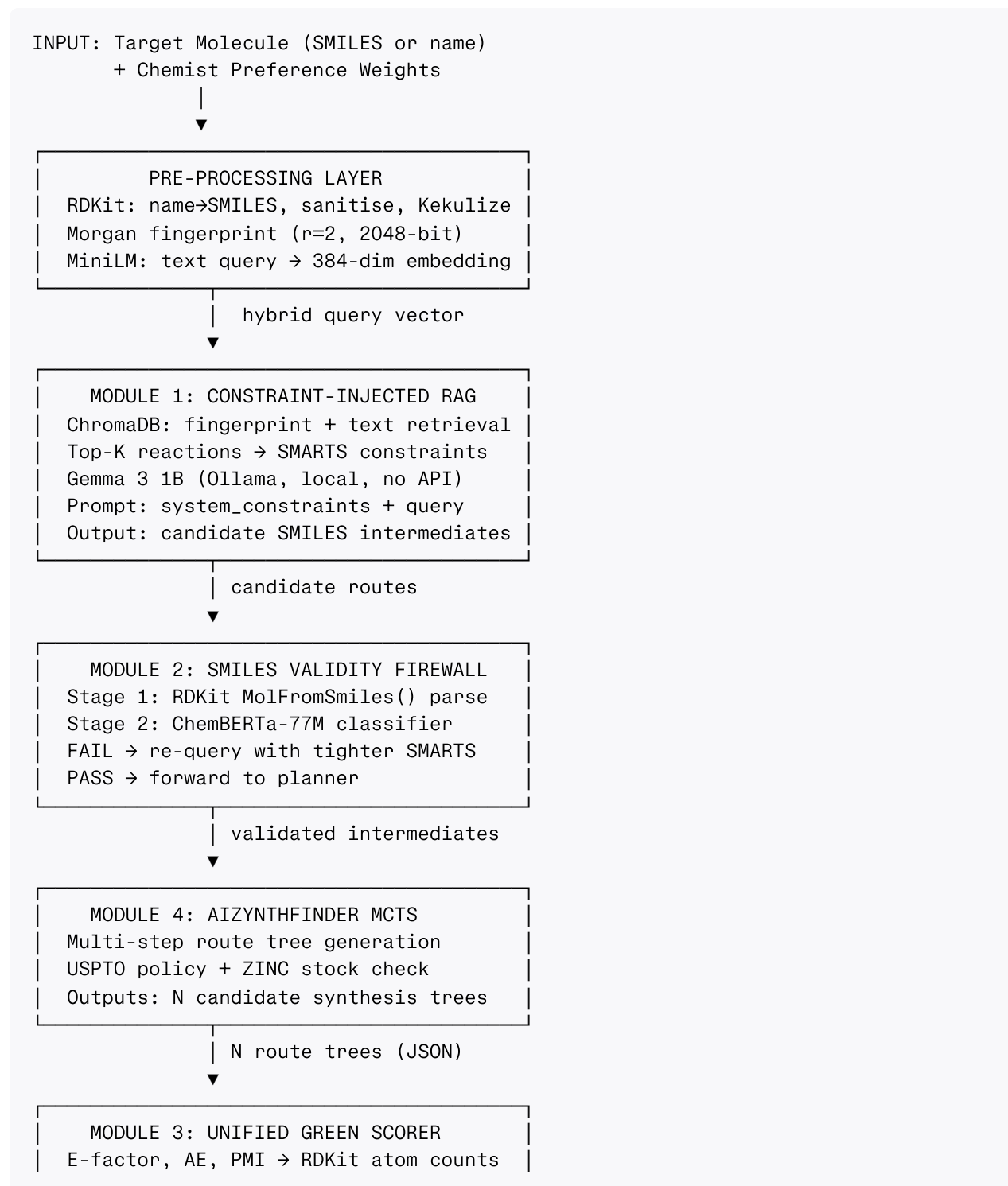
Closes: Gap 5 (static output, no interactivity)

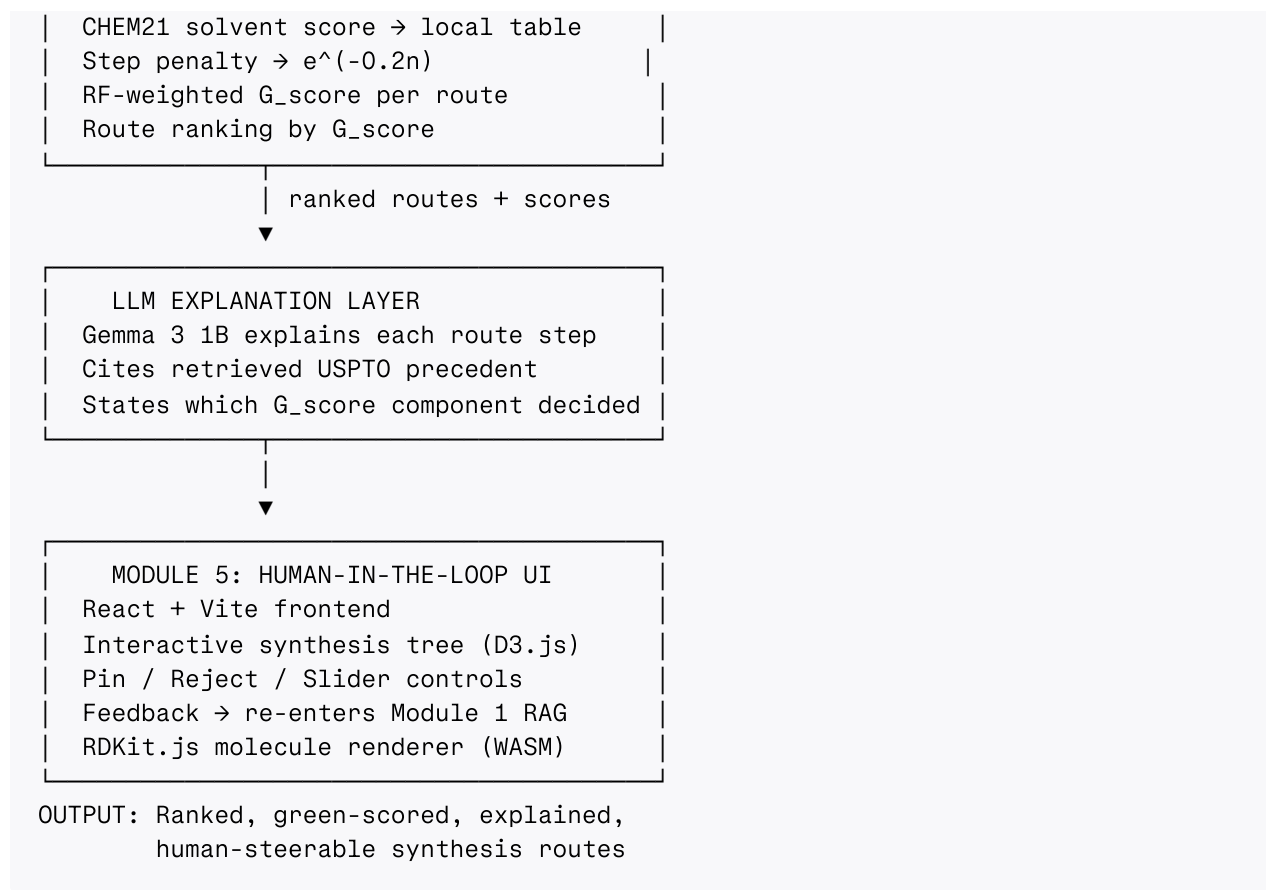
The 2025 NIH paper explicitly identifies this as future work. EcoSynth implements it: [\[17\]](#)

- **Pin a node:** Chemist locks a specific intermediate SMILES — AizynthFinder re-runs MCTS with that node forced into the tree

- **Reject a step:** Remove one reaction from the route → system re-queries RAG with the rejected SMARTS pattern blacklisted
- **Preference sliders:** Three sliders — Greenness weight, Step minimisation weight, Commercial availability weight — update the w_i vector in Module 3 in real-time
- **Explanation panel:** For every step, the LLM explains the reaction in plain English, cites the retrieved precedent reaction from USPTO, and states which green metric was the deciding factor

Full System Architecture





Proof & Accuracy Framework

Each module has a **specific, measurable proof** you can report in a paper:

Module	Experiment	Metric	Expected Result
Module 1 (RAG Constraints)	Compare constrained vs. unconstrained Gemma 3 1B on 200 USPTO test molecules	% chemically valid routes	Constrained: >90%, Unconstrained: ~55% [5]
Module 2 (Firewall)	Evaluate ChemBERTa classifier on held-out hallucinated SMILES test set	F1-score, Precision, Recall	F1 > 0.93 based on ChemBERTa fine-tuning benchmarks [7] [18]
Module 3 (Green Scorer)	Correlate G_score vs. expert greenness rankings from ACS GCI published dataset	Spearman's ρ	$\rho > 0.75$ vs. $\rho < 0.45$ for any single metric [9]
Module 4 (AizynthFinder)	Measure route validity at depth ≤ 4 on USPTO-50k benchmark	Top-1 accuracy	Matches AizynthFinder baseline ~47% [2] , EcoSynth adds green filtering on top
Module 5 (Steering)	User study: routes accepted with vs. without steering	Mean interaction rounds to acceptance	>40% reduction in rounds [17]

What Makes the Paper Publishable

The claim is not "we built an AI for green chemistry" — that already exists. The claim is:

"EcoSynth is the first system to demonstrate that constraint-injected RAG reduces LLM hallucination in retrosynthesis to <10%, that a Random Forest-fused multi-metric green scorer achieves Spearman's $\rho > 0.75$ against expert rankings, and that human-in-the-loop steering reduces synthesis plan refinement effort by >40% — all running on consumer-grade CPU hardware with zero external API dependency."

Every part of that sentence is falsifiable, testable, and directly responsive to a named gap in current literature. That is what separates a research contribution from a project report. ^[16] ^[19]
^[9] ^[17] ^[5]

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1. <https://github.com/MolecularAI/aizynthfinder>
2. <https://molecularai.github.io/aizynthfinder/>
3. <https://www.nature.com/articles/s41467-023-41698-5>
4. <https://arxiv.org/html/2502.17506v3>
5. <https://pubs.acs.org/doi/full/10.1021/acs.jcim.4c02322>
6. <https://rewire.it/blog/chemberta-when-language-models-learned-to-speak-chemistry/>
7. <https://ar5iv.labs.arxiv.org/html/2209.01712>
8. <https://github.com/frogstar-world-b/Transfer-Learning-ChemBERTa>
9. <https://pubmed.ncbi.nlm.nih.gov/40839460/>
10. <https://www.acs.org/green-chemistry-sustainability/research-innovation/research-tools.html>
11. <http://www.chem21.eu/project/chem21-solvent-selection-guide/>
12. <https://chemistryforsustainability.org/tools-metrics/chem21-solvent-selection-guide>
13. [faith-101-1.py](#)
14. <https://chemistry-europe.onlinelibrary.wiley.com/doi/10.1002/cssc.202500809>
15. <https://www.biorxiv.org/content/10.1101/2025.03.07.642029v1.full-text>
16. <https://pubs.acs.org/doi/10.1021/acs.jmedchem.4c02749>
17. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12258405/>
18. <https://pubmed.ncbi.nlm.nih.gov/41534634/>
19. <https://gwern.net/doc/reinforcement-learning/model/alphago/2024-striethkalthoff.pdf>
20. <https://www.sciencedirect.com/science/article/abs/pii/S1359644626000073>
21. <http://www.jatit.org/volumes/Vol103No16/34Vol103No16.pdf>
22. <https://deepchem.io/tutorials/transfer-learning-with-chemberta-transformers/>
23. <https://www.nature.com/articles/s41598-024-76440-8>

24. <https://learning.acsgcipr.org/guides-and-metrics/solvent-selection-guides/the-chem21-solvent-selection-guide/>